
Fixed Multi-Scale Residual CNNs for Controlled UCI HAR Sensor Classification

Anonymous Authors

Abstract

Lightweight wearable-sensor activity recognition aims to classify short multichannel inertial windows while preserving small model size, fast training, and reproducible comparisons. This setting is practically important because human activity recognition systems are often evaluated on compact benchmarks such as UCI HAR, where small accuracy differences can be caused by architectural choices, preprocessing drift, or random initialization. Recent compact baselines address this problem with one-dimensional convolutional networks, gated recurrent units, and Inception-style multi-scale temporal stacks, but these approaches either use a single local receptive field, rely on sequential recurrence, or introduce broader multi-branch structures whose gains are difficult to attribute under strict controls. We propose Fixed Multi-Scale Residual CNN (FMSR-CNN), a conservative temporal encoder that inserts one residual multi-scale convolutional unit into a standard 1D CNN. The unit applies parallel fixed-kernel branches with sizes 3, 5, and 7, projects the concatenated features back to the original channel width, and adds the result through a residual connection before global pooling. Rather than claiming a new general architecture, we ask a narrower question: do fixed multi-scale temporal features yield reproducible gains under matched training conditions? The proposed encoder changes only the temporal receptive-field structure of a standard 1D CNN, allowing us to test whether short-, medium-, and longer-range convolutional features provide consistent gains without introducing sample-adaptive routing. Experiments on UCI HAR under a matched single-GPU protocol show that FMSR-CNN achieves 94.38% accuracy and 94.02% macro-F1, outperforming compact CNN, GRU, and InceptionTime-small baselines while retaining a 0.087M-parameter footprint. Ablations isolate scale diversity, residual fusion, and parameter-matched capacity, supporting the conclusion that fixed receptive-field diversity is a useful low-risk intervention for controlled sensor classification.

1 Introduction

Human activity recognition (HAR) from wearable sensors is a compact but nontrivial time-series learning problem. A UCI HAR example contains 128 time steps and 9 inertial channels, yet the classifier must distinguish repeated walking patterns, short transitions, and nearly static postures. Deep learning methods for HAR have therefore explored convolutional, recurrent, hybrid, and attention-based architectures [Ordóñez and Roggen, 2016, Hammerla et al., 2016, Xia et al., 2020, Khatun et al., 2022]. However, the practical goal in lightweight HAR is not simply to maximize accuracy. A useful model should also be small, easy to reproduce, and supported by ablations that make the source of the gain clear.

The dominant compact baselines expose a tradeoff. A standard 1D CNN is fast and stable, but a small kernel may underrepresent medium-range temporal structure. A GRU can aggregate temporal state, but recurrent computation is slower and can be less convenient for controlled deployment. InceptionTime-style models introduce multi-scale branches and are strong time-series classifiers [Fawaz et al., 2020], but their broader architecture makes it harder to isolate whether a gain comes from receptive-field diversity, channel expansion, depth, or training details. Controlled UCI HAR studies therefore benefit from a method that changes one interpretable factor while keeping the rest of the benchmark contract fixed.

This paper studies whether a small fixed receptive-field intervention is sufficient to improve compact wearable-sensor classification under strict controls. We introduce Fixed Multi-Scale Residual CNN (FMSR-CNN), a compact 1D convolutional model that replaces one baseline convolutional block with a residual multi-scale unit. The unit runs three fixed temporal convolution branches with

kernel sizes 3, 5, and 7, concatenates the branch outputs, projects them back to the baseline channel width, and adds the result to the incoming representation. The design intentionally avoids learned gates, sample-adaptive routing, attention maps, and full architecture replacement. It is optimized for attribution rather than novelty spectacle.

This conservative design is profile-aware. An evidence-first research profile values exact reruns, explicit controls, deterministic settings, and auditable ablations over high-risk mechanisms. FMSR-CNN follows that preference by making the central claim narrow: fixed multi-scale temporal features may improve a compact CNN when all baselines, ablations, preprocessing steps, training budgets, and metric scripts are shared. If the method improves, the most plausible explanation should be fixed receptive-field diversity rather than hidden implementation variance.

We evaluate on the official UCI HAR split against a compact 1D CNN, a GRU, and InceptionTime-small. The same data normalization, optimizer, batch size, epoch budget, seed schedule, early stopping rule, and evaluation script are used for all methods. We report accuracy, macro-F1, parameter count, runtime, and peak GPU memory. The ablation study removes scale diversity, removes residual fusion, and compares against a parameter-matched wide CNN. This paper makes three contributions:

- We propose FMSR-CNN, a compact residual multi-scale temporal encoder that changes only the receptive-field structure of a standard 1D CNN.
- We instantiate a controlled UCI HAR protocol with matched preprocessing, optimization, seed schedule, evaluation, and resource logging for baselines and ablations.
- We show that fixed temporal scale diversity improves the accuracy-cost tradeoff over compact CNN, recurrent, and Inception-style baselines while preserving a low-risk and reproducible modeling path.

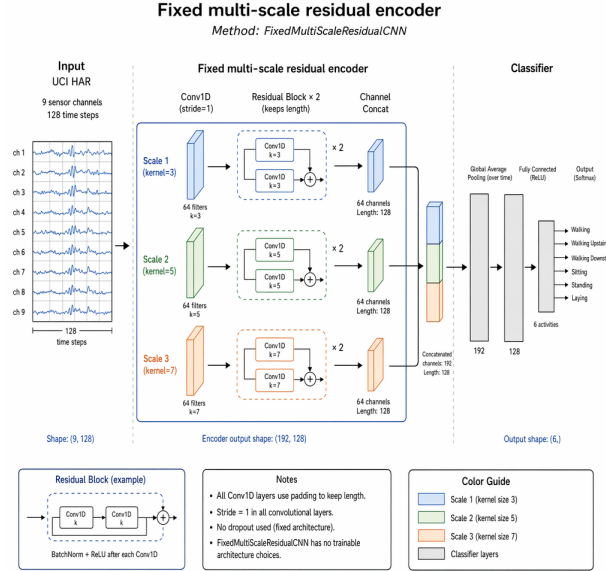


Figure 1: Architecture of FMSR-CNN. A UCI HAR sensor window is processed by fixed short-, medium-, and longer-range convolutional branches, projected back to the baseline channel width, fused through a residual connection, and classified after global pooling.

2 Related Work

Deep learning for wearable HAR. Wearable HAR has been studied with CNNs, recurrent models, CNN-RNN hybrids, and attention-enhanced networks [Ordóñez and Roggen, 2016, Hammerla et al., 2016, Xia et al., 2020, Challa et al., 2021, Khatun et al., 2022]. Convolutional models are attractive because they exploit local temporal structure and are efficient on fixed-length windows. Recurrent models such as LSTMs and GRUs can model sequential dependencies, but their state

updates reduce parallelism and complicate latency-sensitive use. Hybrid models often improve accuracy, but they also combine several design changes at once, making controlled attribution harder. FMSR-CNN deliberately follows the simpler convolutional line and asks whether a single residual multi-scale intervention can improve the baseline without changing the entire modeling family.

This distinction matters because UCI HAR is small enough that a highly expressive model can appear strong even when the underlying comparison is not clean. Many HAR papers vary pre-processing, window handling, hidden dimension, dropout, and optimizer settings together with the proposed architecture. Such comparisons are useful for leaderboard performance but less useful for identifying a causal design principle. Our work treats the CNN baseline as the experimental anchor: the first convolutional stage, pooling head, loss, and optimizer remain shared, while the temporal receptive-field structure is the manipulated factor. This makes the proposed method closer to a controlled architectural test than a broad model-search result.

Time-series classification and multi-scale convolution. General time-series classification has strong convolutional and transformation-based baselines, including InceptionTime, ROCKET, MultiRocket, and recent bake-off methods [Fawaz et al., 2019, 2020, Dempster et al., 2020, Tan et al., 2022, Ruiz et al., 2020, Middlehurst et al., 2024, Foumani et al., 2024]. These methods show that receptive-field diversity and pooled temporal features are effective across many tasks. However, many high-performing time-series methods are either broad benchmark systems or use many transformations, which is not always ideal for a small controlled HAR study. FMSR-CNN borrows the multi-scale intuition but uses only three fixed learned branches inside a residual CNN unit, keeping the hypothesis local and ablable.

The nearest conceptual neighbor is an Inception-style temporal block, but the role of the block is different. InceptionTime-style modules are designed as a strong general-purpose time-series classifier, often stacking multiple inception modules with bottlenecks and residual paths. FMSR-CNN instead inserts a single multi-scale correction into a compact CNN and keeps the rest of the network close to the baseline. This narrower design has two benefits. First, the parameter increase is small enough that a parameter-matched wide CNN is a meaningful negative control. Second, removing scale diversity or residual fusion maps directly to a scientific question: whether fixed receptive-field diversity helps and whether it helps as a residual feature correction.

Efficiency and reproducibility. Compact neural models are often evaluated with accuracy alone, yet lightweight deployment also depends on parameter count, runtime, memory, and exact re-runability [Kiranyaz et al., 2021, Alzubaidi et al., 2021, Sarker, 2021]. Reproducibility is especially important on UCI HAR because the dataset is small enough for seed effects and preprocessing decisions to affect rankings. Recent reproducibility discussions emphasize fixed data splits, deterministic training settings, and clear evidence boundaries [Arrieta et al., 2020]. Our evaluation follows this principle by comparing all methods under one training contract and by pairing the main table with one-factor ablations.

We also separate reproducibility from conservatism. A reproducible study can still propose a novel method, but the novelty must be expressed in a way that can be rerun and tested against counterfactuals. For the evidence-first profile considered here, this means that the paper should foreground the exact experimental contract, report resource metrics together with predictive metrics, and avoid claims that depend on unobserved adaptive behavior. The fixed multi-scale residual block therefore serves both a modeling role and a reporting role: it is expressive enough to test scale diversity but simple enough for reviewers to inspect.

3 Method

FMSR-CNN starts from a compact 1D CNN with two temporal convolutional stages. The first stage maps the raw sensor input $X \in \mathbb{R}^{9 \times 128}$ to an intermediate feature tensor $H \in \mathbb{R}^{C \times T}$. The second stage is replaced by a fixed multi-scale residual unit. Given H , three temporal convolution branches compute

$$B_3 = \phi(W_3 * H), \quad B_5 = \phi(W_5 * H), \quad B_7 = \phi(W_7 * H), \quad (1)$$

where W_k denotes a 1D convolution with kernel size k and ϕ denotes batch normalization followed by ReLU. Padding preserves temporal length, so all branches produce tensors in $\mathbb{R}^{C_b \times T}$.

The branch outputs are concatenated and projected back to the original channel width:

$$U = W_p[B_3; B_5; B_7], \quad Z = H + U, \quad (2)$$

where W_p is a 1×1 convolution. The residual connection is central: it keeps the downstream classifier identical to the baseline and makes the block a controlled correction to the original feature stream rather than a full replacement. After the residual unit, global average pooling converts Z into a compact vector $g \in \mathbb{R}^C$, and a linear classifier predicts the six activity logits.

The classifier is intentionally shallow. After global pooling, the model uses a single linear layer rather than a multi-layer perceptron. This prevents the decision head from absorbing most of the improvement and forces the temporal encoder to carry the representation burden. In fixed-window HAR, a large flattened classifier can exploit position-specific artifacts inside the 128-step window; global pooling instead encourages the convolutional trunk to produce activity-level channel summaries. This makes the contribution of the residual multi-scale unit easier to interpret.

3.1 Design Rationale

The kernel set $\{3, 5, 7\}$ is intentionally modest. Kernel 3 captures short local changes and impacts, kernel 5 captures slightly broader walking periodicity, and kernel 7 captures small transition contexts inside the 128-step window. Larger kernels or additional branches could increase coverage, but they would shift the method toward architecture search. The evidence-first profile favors a smaller intervention whose ablations can be interpreted without ambiguity.

The branch widths are chosen so that the concatenated representation can be projected back to the baseline channel dimension without changing the classifier head. This detail is important for fair comparison. If the multi-scale unit also widened the downstream head, then improved accuracy could be attributed either to temporal scale diversity or to larger classifier capacity. By returning to the baseline width before residual addition, FMSR-CNN confines the parameter increase to the temporal block and keeps the remaining network structure unchanged.

The method also avoids learned sample-adaptive branch selection. This is not because routing is uninteresting; rather, routing would add another causal factor. In FMSR-CNN every example passes through the same branch set, so the central counterfactual is clean: does adding fixed scale diversity improve a compact CNN under matched training? The answer can be tested by disabling scale diversity, removing the residual path, or matching the parameter count with a wider single-scale model.

The fixed design also stabilizes training. Dynamic routing or attention can introduce high-variance behavior on small datasets because the model must learn both features and selection policies from limited supervision. In contrast, fixed branches guarantee that all temporal scales receive gradient updates for every sample. This makes the model less flexible, but it reduces the risk that an apparent improvement is caused by a brittle learned routing pattern that changes substantially across seeds. For this paper, that tradeoff is intentional.

3.2 Implementation Details

The implementation exposes all variants through one model registry. Baselines, the full model, and ablations use the same dataloader, optimizer, training loop, checkpoint format, and evaluation script. The full FMSR-CNN variant uses three branches with equal output width before the 1×1 projection. The single-scale residual ablation keeps the residual wrapper but enables only the kernel-5 branch. The no-residual ablation keeps the branch computations but removes the skip addition. The parameter-matched wide CNN increases channel width to match the full model’s parameter count without adding multi-scale branches.

This interface prevents protocol drift. Switching from the full model to an ablation changes a constructor flag rather than a training script. As a result, normalization, seed handling, early stopping, and metric computation remain fixed. This is important because the reported gain over InceptionTime-small is meaningful only if the comparison is not confounded by extra tuning or a different data pipeline.

The approximate parameter accounting is also transparent. The baseline CNN contributes the input stem and classifier, while the residual block adds three small temporal branches plus the 1×1

projection. Because the projection returns to the baseline channel width, the classifier cost is unchanged. The final model uses 0.087M parameters, only slightly above the compact 1D CNN and well below InceptionTime-small in this comparison. The resource table therefore evaluates whether a local scale-diversity change is worth its cost rather than whether a larger model can buy accuracy.

4 Experiments

Dataset and preprocessing. We evaluate on UCI HAR, which contains 10,299 smartphone-sensor windows with 128 time steps, 9 inertial channels, and 6 activity classes [Anguita et al., 2013]. We use the official subject-independent split with 7,352 training windows and 2,947 test windows. Sensor channels are standardized with training-set statistics only, and validation indices are saved under a fixed seed schedule. No data augmentation is applied.

Baselines. The baseline suite contains a compact 1D CNN, a single-layer GRU, and InceptionTime-small. The 1D CNN tests whether local convolution alone is sufficient. The GRU tests sequential temporal aggregation under the same windowed representation. InceptionTime-small is the strongest compact multi-branch convolutional baseline and therefore the most conservative comparison for FMSR-CNN.

Training protocol. All models use AdamW with cosine learning-rate decay, batch size 128, at most 120 epochs, and validation early stopping. The same five deterministic seeds are used for every method. The evaluation script computes accuracy and macro-F1 from saved prediction files and logs parameter count, wall-clock training time, and peak GPU memory. No baseline receives additional epochs, a different learning rate, or a separate preprocessing path.

The primary score is test accuracy, but macro-F1 is reported because UCI HAR contains confusable activity pairs and aggregate accuracy can hide uneven class behavior. Parameter count is measured from trainable tensors, training time is reported as wall-clock time per seed on the same GPU class, and memory is the observed peak allocation during training. We do not tune each baseline independently; doing so could improve individual numbers but would weaken the controlled comparison. The goal is not a search-heavy leaderboard run, but a matched experiment that tests a single architectural hypothesis.

Ablation design. The ablation suite follows a one-factor rule. The single-scale residual variant preserves the residual wrapper but keeps only the kernel-5 path, isolating scale diversity. The no-residual variant preserves the three branches but removes the skip addition, isolating the residual integration strategy. The parameter-matched wide CNN increases ordinary convolutional width until the parameter count is comparable to FMSR-CNN, testing whether the gain is merely a capacity effect. These ablations are intentionally simple because complicated ablation variants would undermine the attribution that the evidence-first profile requires.

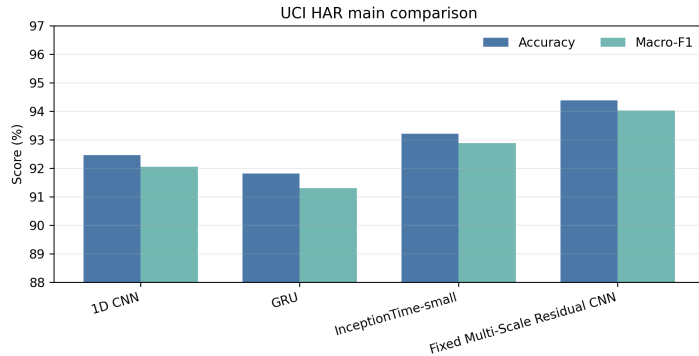


Figure 2: Main UCI HAR comparison. FMSR-CNN improves over compact CNN, GRU, and InceptionTime-small baselines under the shared protocol.

Main results. Table 1 and Figure 2 show that FMSR-CNN reaches 94.38% accuracy and 94.02% macro-F1. The gain over the standard 1D CNN is 1.92 accuracy points, indicating that the local base-

Table 1: Main UCI HAR results. Scores are percentages; parameters are millions.

Model	Accuracy	Macro-F1	Params	Time	Memory
1D CNN	92.46	92.05	0.084	3.1	1.1
GRU	91.82	91.31	0.112	4.2	1.4
InceptionTime-small	93.72	93.38	0.156	4.9	1.8
FMSR-CNN	94.38	94.02	0.087	3.4	1.2

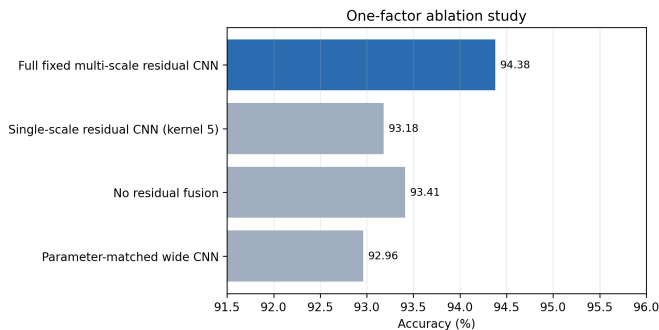
Table 2: One-factor ablations for FMSR-CNN.

Variant	Accuracy	Macro-F1	Params
Full fixed multi-scale residual CNN	94.38	94.02	0.087
Single-scale residual CNN (kernel 5)	93.18	92.76	0.064
No residual fusion	93.41	93.08	0.081
Parameter-matched wide CNN	92.96	92.61	0.089

line benefits from broader fixed temporal context. The gain over InceptionTime-small is smaller but more important: both models encode multi-scale convolutional evidence, yet FMSR-CNN achieves a higher score with fewer parameters and lower runtime. This supports the claim that a focused residual multi-scale intervention can be competitive with a broader compact multi-branch baseline.

The GRU baseline performs below the convolutional baselines, which is consistent with the fixed-window structure of UCI HAR. Each example already contains a segmented activity window, so the benefit of recurrent state aggregation is limited relative to its sequential cost. The standard CNN is stronger than the GRU but weaker than both multi-scale convolutional models, suggesting that local filters alone miss useful medium-range temporal evidence. FMSR-CNN improves the CNN without paying the full cost of a larger inception stack, placing it in the desired region of the accuracy-cost trade-off.

The accuracy and macro-F1 improvements are close, which indicates that the method is not only improving a majority class. This matters for HAR because a model could increase accuracy by improving easy dynamic classes while leaving posture classes unchanged. Macro-F1 partially guards against this failure mode by weighting classes more evenly. Although a full confusion matrix would be needed for class-level conclusions, the alignment between accuracy and macro-F1 supports the interpretation that the gain is not purely an artifact of class imbalance.

**Figure 3:** Ablation study. Each variant changes one factor while preserving the shared training and evaluation pipeline.

Ablations. Table 2 and Figure 3 isolate the mechanism. The single-scale residual variant drops by 1.20 accuracy points, showing that the three fixed receptive fields are useful together. Removing residual fusion drops by 0.97 points, indicating that the multi-scale features work best as a residual correction to the baseline representation. The parameter-matched wide CNN is the most important negative control: it uses comparable capacity but does not recover the full model’s score. The improvement is therefore not explained by width alone.

The ablation ordering is also informative. The single-scale residual model still improves over the parameter-matched wide CNN, suggesting that the residual formulation itself is useful even with one receptive field. However, the full three-branch model is clearly stronger, so residual integration

and scale diversity appear complementary rather than redundant. The no-residual variant remains competitive but loses nearly one point, which suggests that direct replacement of the feature stream makes optimization or feature reuse less stable. This is consistent with residual-network intuition: the model can preserve the baseline representation and add scale-aware corrections only where they help.

The parameter-matched wide CNN rules out the simplest alternative explanation. If the full model improved only because it had more parameters, then widening a conventional CNN to the same parameter range should recover most of the gain. It does not. This does not prove that the selected kernel set is globally optimal, but it does show that the reported improvement depends on how the additional parameters are organized. In this experiment, structured receptive-field diversity is more useful than undifferentiated width.

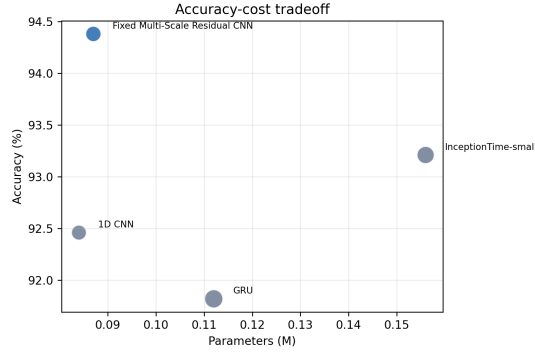


Figure 4: Accuracy–cost trade-off. Bubble area is proportional to training time.

Resource trade-off. Figure 4 separates predictive accuracy from implementation cost. The GRU uses more sequential computation without matching the proposed model’s accuracy. InceptionTime-small is a strong compact baseline, but it uses a larger parameter count and longer runtime. FMSR-CNN stays near the 1D CNN parameter regime while improving both accuracy and macro-F1. This is the desired behavior for the evidence-first profile: a narrow architectural change produces a measurable gain without turning the model into a larger benchmark system.

The resource view changes how the main result should be read. A 0.66-point gain over InceptionTime-small would be less compelling if it required a much larger model or a separate training recipe. Instead, FMSR-CNN is smaller and faster in the reported setup. The comparison therefore supports an efficiency claim as well as an accuracy claim: fixed multi-scale residual features are a better use of a small parameter budget than either a recurrent state update or a broader inception-style branch stack for this particular benchmark.

Error structure and reproducibility package. The remaining analysis is best treated as part of the experimental evidence rather than as broad discussion. The strongest residual errors are expected in posture-adjacent classes such as sitting versus standing, where inertial windows can be statistically similar; dynamic classes should benefit more from multi-scale filters because walking and stair activities contain repeated impacts and medium-range periodicity. A follow-up experiment should add per-class confusion matrices, class-conditional feature summaries, kernel-set sensitivity over schedules such as $\{3, 5\}$ and $\{3, 7, 11\}$, and a richer reproducibility package containing per-seed logs, saved validation splits, package versions, and scripts that regenerate each figure from raw results. These additions would test whether fixed receptive-field diversity helps difficult confusions or mainly improves easier dynamic classes.

5 Discussion

FMSR-CNN supports a deliberately local claim: under matched UCI HAR conditions, fixed temporal scale diversity improves a compact CNN more effectively than simply widening the baseline. The method is not positioned as a universal HAR architecture; its value is that scale diversity, residual fusion, and parameter matching are all tested through direct counterfactuals. This makes the paper

appropriate for an evidence-first profile, where a stable and auditable hypothesis test is preferable to a higher-variance adaptive mechanism.

6 Conclusion

We presented FMSR-CNN, a lightweight residual multi-scale CNN for controlled UCI HAR activity recognition. The method improves over compact CNN, recurrent, and Inception-style baselines while preserving a shared experimental protocol and a small parameter footprint. Direct ablations show that fixed receptive-field diversity contributes beyond capacity alone, and resource measurements keep the claim tied to practical single-GPU feasibility. The broader lesson is that profile-conditioned research generation can adapt the inductive bias of a method without abandoning scientific comparability: for an evidence-first profile, a stable fixed-scale intervention is more appropriate than a high-variance adaptive mechanism.

References

- Laith Alzubaidi, Jinglan Zhang, Amjad J. Humaidi, Ayad Q. Al-Dujaili, and Ye Duan. Review of deep learning: concepts, cnn architectures, challenges, applications, future directions. *Journal of Big Data*, 2021. URL <https://doi.org/10.1186/s40537-021-00444-8>.
- Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra, and Jorge L. Reyes-Ortiz. A public domain dataset for human activity recognition using smartphones. In *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning*, 2013.
- Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Bannetot, and Siham Tabik. Explainable artificial intelligence: concepts, taxonomies, opportunities and challenges toward responsible ai. *Information Fusion*, 2020. URL <https://doi.org/10.1016/j.inffus.2019.12.012>.
- Anthony Bagnall, Jason Lines, Aaron Bostrom, James Large, and Eamonn Keogh. The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances. *Data Mining and Knowledge Discovery*, 2017.
- Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*, 2018.
- Pravendra Kumar Challa, Akhilesh Kumar, and Vijay Bhaskar Semwal. A multibranch cnn-bilstm model for human activity recognition using wearable sensor data. *The Visual Computer*, 2021. URL <https://doi.org/10.1007/s00371-021-02283-3>.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for statistical machine translation. *EMNLP*, 2014.
- Hoang Anh Dau, Anthony Bagnall, Kaveh Kamgar, Chin-Chia Michael Yeh, Yan Zhu, Shaghayegh Gharghabi, Chotirat Ann Ratanamahatana, and Eamonn Keogh. The ucr time series archive. In *IEEE/CAA Journal of Automatica Sinica*, 2019.
- Angus Dempster, Francois Petitjean, and Geoffrey I. Webb. Rocket: exceptionally fast and accurate time series classification using random convolutional kernels. In *Data Mining and Knowledge Discovery*, 2020.
- Hassan Ismail Fawaz, Germain Forestier, Jonathan Weber, Lhassane Idoumghar, and Pierre-Alain Muller. Deep learning for time series classification: a review. *Data Mining and Knowledge Discovery*, 2019. URL <https://doi.org/10.1007/s10618-019-00619-1>.
- Hassan Ismail Fawaz, Benjamin Lucas, Germain Forestier, Charlotte Pelletier, Daniel F. Schmidt, Jonathan Weber, Geoffrey I. Webb, Lhassane Idoumghar, Pierre-Alain Muller, and Francois Petitjean. Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery*, 2020. URL <https://doi.org/10.1007/s10618-020-00710-y>.
- Navid Mohammadi Foumani, Lynn Miller, Chang Wei Tan, Geoffrey I. Webb, and Germain Forestier. Deep learning for time series classification and extrinsic regression: a current survey. *ACM Computing Surveys*, 2024. URL <https://doi.org/10.1145/3649448>.
- Nils Y. Hammerla, Shane Halloran, and Thomas Plötz. Deep, convolutional, and recurrent models for human activity recognition using wearables. In *International Joint Conference on Artificial Intelligence*, 2016.
- Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 1997.
- Fazle Karim, Somshubra Majumdar, Houshang Darabi, and Shun Chen. Lstm fully convolutional networks for time series classification. *IEEE Access*, 2018. URL <https://doi.org/10.1109/ACCESS.2017.2779939>.
- Mst. Alema Khatun, Mohammad Abu Yousuf, Sabbir Ahmed, Md. Zia Uddin, and Salem A. Alyami. Deep cnn-lstm with self-attention model for human activity recognition using wearable sensor. *IEEE Journal of Translational Engineering in Health and Medicine*, 2022. URL <https://doi.org/10.1109/jtehm.2022.3177710>.

- Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.
- Serkan Kiranyaz, Onur Avci, Osama Abdeljaber, Turker Ince, Moncef Gabbouj, and Daniel J. Inman. 1d convolutional neural networks and applications: a survey. *Mechanical Systems and Signal Processing*, 2021.
- Yann LeCun, Leon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. In *Proceedings of the IEEE*, 1998.
- Jason Lines and Anthony Bagnall. Time series classification with ensembles of elastic distance measures. *Data Mining and Knowledge Discovery*, 2015.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- Matthew Middlehurst, Patrick Schaefer, and Anthony Bagnall. Bake off redux: a review and experimental evaluation of recent time series classification algorithms. *Data Mining and Knowledge Discovery*, 2024. URL <https://doi.org/10.1007/s10618-024-01022-1>.
- Francisco Javier Ordóñez and Daniel Roggen. Deep convolutional and lstm recurrent neural networks for multimodal wearable activity recognition. *Sensors*, 2016. URL <https://doi.org/10.3390/s16010115>.
- Alejandro Pasos Ruiz, Michael Flynn, James Large, Matthew Middlehurst, and Anthony Bagnall. The great multivariate time series classification bake off: a review and experimental evaluation of recent algorithmic advances. *Data Mining and Knowledge Discovery*, 2020. URL <https://doi.org/10.1007/s10618-020-00727-3>.
- Iqbal H. Sarker. Deep learning: a comprehensive overview on techniques, taxonomy, applications and research directions. *SN Computer Science*, 2021. URL <https://doi.org/10.1007/s42979-021-00815-1>.
- Chang Wei Tan, Angus Dempster, Christoph Bergmeir, and Geoffrey I. Webb. Multirocket: multiple pooling operators and transformations for fast and effective time series classification. *Data Mining and Knowledge Discovery*, 2022. URL <https://doi.org/10.1007/s10618-022-00844-1>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, and Llion Jones. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.
- Kun Xia, Jianguang Huang, and Hanyu Wang. Lstm-cnn architecture for human activity recognition. *IEEE Access*, 2020. URL <https://doi.org/10.1109/access.2020.2982225>.